Problem 13.3.1

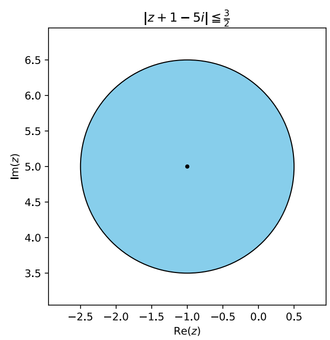
Determine and sketch or graph the sets in the complex plane given by

$$|z + 1 - 5i| \leqq \frac{3}{2}$$

Solution

$$|z - (-1 + 5i)| \leqq \frac{3}{2}$$

This is a closed disc centered at $(-1 + 5i)$ with radius 1.5



Problem 13.3.3

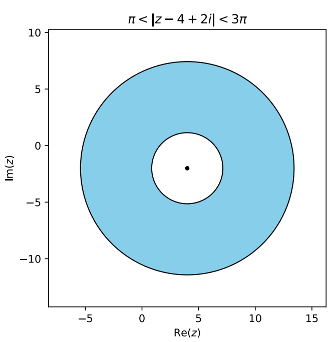
Determine and sketch or graph the sets in the complex plane given by

$$\pi < |z - 4 + 2i| < 3\pi$$

Solution

$$\pi < |z - (4 - 2i)| < 3\pi$$

This is an open disc centered at $(4 - 2i)$ with outer radius 3π, and inner radius π



Problem 13.3.5

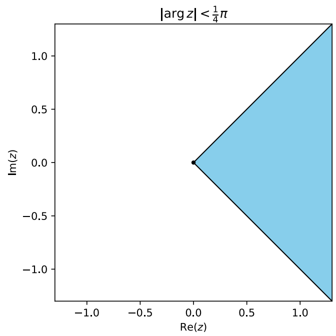
Determine and sketch or graph the sets in the complex plane given by

$$|\arg z| < \frac{1}{4}\pi$$

Solution

$$-\frac{1}{4}\pi < \arg z < \frac{1}{4}\pi$$

This is a wedge from $-\frac{1}{4}\pi$ to $\frac{1}{4}\pi$. There is no bound to the radius of the wedge.



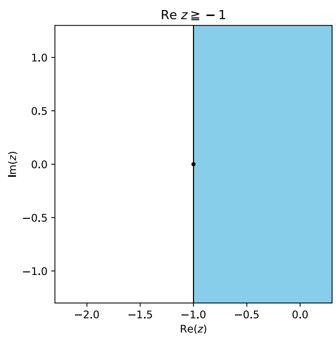
Problem 13.3.7

Determine and sketch or graph the sets in the complex plane given by

$$\operatorname{Re} z \geqq -1$$

Solution

This covers the right half of plane bounded at the left by $x = -1$.



Problem 13.3.11

Find $\operatorname{Re} f$, and $\operatorname{Im} f$ and their values at the given point z.

$$f(z) = \frac{1}{1-z} \text{ at } 1 - i$$

Solution

$z = x + iy$

$$\begin{aligned}
f(z) &= \frac{1}{1-(x+iy)} \\
&= \frac{1}{(1-x)-iy} \\
&= \frac{1}{(1-x)-iy} \cdot \frac{(1-x)+iy}{(1-x)+iy} \\
&= \frac{(1-x)+iy}{(1-x)^2+y^2}
\end{aligned}$$

$$\boxed{\operatorname{Re} f(z) = \frac{1-x}{(1-x)^2+y^2}} \quad \boxed{\operatorname{Im} f(z) = \frac{y}{(1-x)^2+y^2}}$$

$$\boxed{\operatorname{Re} f(1-i) = 0} \quad \boxed{\operatorname{Im} f(1-i) = -1}$$

Problem 13.3.23

Find the value of the derivative of

$$\frac{z^3}{(z+1)^3} \text{ at } i$$

Solution

$$f(z) = \left(\frac{z}{z+1}\right)^3$$

$$\begin{aligned}
f'(z) &= 3\left(\frac{z}{z+1}\right)^2 \left(\frac{(z+1)-z}{(z+1)^2}\right) \\
&= \frac{3z^2}{(z+1)^4}
\end{aligned}$$

$$\begin{aligned}
f'(i) &= \frac{3i^2}{(i+1)^4} \\
&= \boxed{\frac{3}{4}}
\end{aligned}$$

Problem 13.4.3

Is f analytic?

$$f(z) = e^{-2x}(\cos 2y - i \sin 2y)$$

Solution

$$f(z) = u(x,y) + iv(x,y)$$

$$\begin{aligned} u &= e^{-2x}\cos 2y & v &= -e^{-2x}\sin 2y \\ u_x &= -2e^{-2x}\cos 2y & v_x &= 2e^{-2x}\sin 2y \\ u_y &= -2e^{-2x}\sin 2y & v_y &= -2e^{-2x}\cos 2y \end{aligned}$$

Since $u_x = v_y$ and $u_y = -v_x$, f is **analytic**.

Problem 13.4.5

Is f analytic?

$$f(z) = \mathrm{Re}(z^2) - i\,\mathrm{Im}(z^2)$$

Solution

$$z^2 = (x+iy)^2 = x^2 - y^2 + i2xy$$

$$f(z) = u(x,y) + iv(x,y)$$

$$\begin{aligned} u &= x^2 - y^2 & v &= 2xy \\ u_x &= 2x & v_x &= 2y \\ u_y &= -2y & v_y &= 2z \end{aligned}$$

Since $u_x = v_y$ and $u_y = -v_x$, f is **analytic**.

Problem 13.4.7

Is f analytic?

$$f(z) = \frac{i}{z^8}$$

Solution

$$f'(z) = -8\frac{i}{z^9}$$

Since f is differentiable everywhere (except $z = 0$), f is **analytic**.

Problem 13.4.13

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x,y) + iv(x,y)$.

$$u = xy$$

Solution

$\Delta\, u = 0$, and therefore the function is **harmonic**.

$$u_x = y = v_y \qquad \Longrightarrow \qquad v = \frac{1}{2}y^2 + h(x)$$

$$v_x = h'(x) = -u_y = x \qquad \Longrightarrow \qquad h(x) = \frac{1}{2}x^2 + c$$

Therefore,

$$f(z) = xy + i\frac{1}{2}(y^2 + x^2 + c)$$

Problem 13.4.17

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

$$v = (2x + 1)y$$

Solution

Problem 13.4.23

Determine a and b so that the given function is harmonic and find a harmonic conjugate.

$$u = \cosh ax \cos y$$

Solution

Problem 13.5.3

Find e^z in the form $u + iv$ and $|e^z|$ if

$$z = 2\pi i(1 + i)$$

Solution

Problem 13.5.9

Write in exponential form

$$4 + 3i$$

Solution

Problem 13.5.15

Fine Re and Im of

$$\exp(z^2)$$

Solution

Problem 13.5.17

Fine Re and Im of

$$\exp(z^3)$$

Solution

Problem 13.5.19

Find all solutions and graph some of them in the complex plane.

$$e^z = 1$$

Solution